

Intel Ice Lake and Tiger Lake

List of instruction timings and μ op breakdown

Explanation of column headings:

Instruction:	Name of instruction. Multiple names mean that these instructions have the same data. Instructions with or without V name prefix behave the same unless otherwise noted.
Operands:	i = immediate data, r = register, mm = 64 bit mmx register, x = 128 bit xmm register, y = 256 bit ymm register, z = 512 bit zmm register, xy = xmm or ymm register, v = any vector register (mmx, xmm, ymm, zmm). m = memory operand, m32 = 32-bit memory operand, etc.
μops fused domain:	The number of μ ops at the decode, rename, allocate and retire stages in the pipeline. Fused μ ops count as one.
μops unfused domain:	The total number of μ ops for all execution ports. Fused μ ops count as two. Fused macro-ops count as one. The instruction has μ op fusion if this number is higher than the number under fused domain. Some operations are not counted here if they do not go to any execution port or if the counters are inaccurate.
μops each port:	The number of μ ops for each execution port. p0 means a μ op to execution port 0. p01 means a μ op that can go to either port 0 or port 1. p0 p1 means two μ ops going to port 0 and 1, respectively. Port 0: Integer, f.p. and 256 bit vector ALU, mul, div, branch Port 1: Integer, f.p. and 256 bit vector ALU (re-routed to port 0 for 512-bit vectors) Port 23: Load Port 49: Store Port 78: Store address Port 5: Integer and 512 bit vector ALU Port 6: Integer ALU, branch
Latency:	This is the delay that the instruction generates in a dependency chain. The numbers are minimum values. Cache misses, misalignment, and exceptions may increase the clock counts considerably. Where hyperthreading is enabled, the use of the same execution units in the other thread leads to inferior performance. Subnormal numbers may increase the latency. The time unit used is core clock cycles, not the reference clock cycles given by the time stamp counter.
Reciprocal throughput:	The average number of core clock cycles per instruction for a series of independent instructions of the same kind in the same thread.

Integer instructions

Instruction	Operands	μ ops fused domain	μ ops unfused domain	μ ops each port	Latency	Reciprocal throughput	Comments
Move instructions							
MOV	r,i	1	1	p0156		0.25	all addressing modes
MOV	r,r	1	1	p0156	1	0.25	
MOV	r8,r8h r8h,r8	1	1	p0156	2	0.25	
MOV	r8/16,m	1	2	p23 p0156		0.5	
MOV	r32/64,m	1	1	p23	3	0.5	
MOV	m,r	1	2	p49 p78	2	1	
MOV	m,i	1	2	p49 p78		1	
MOVNTI	m,r	2	2	p23 p4	~400	~30	
MOVSX MOVZX	r,r	1	1	p0156	1	0.25	
MOVSD							
MOVSX MOVZX	r16,m8	1	2	p23 p0156	3	0.5	

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MOVSX MOVZX MOVSLD	r,m	1	1	p23	3	0.5	all other combinations
CMOVcc	r,r	1	1	p06	1	0.5	
CMOVcc	r,m	1	2	p06 p23		0.5	
XCHG	r,r	3	3	3p0156	1-2	1	
XCHG	r,m	8	8		19		implicit lock
XLAT		3	3	p23 2p0156	7	1	
PUSH	r	1	2	p49 p78	3	0.5	
PUSH	i	1	2	p49 p78		0.5	
PUSH	m	3	3	p23 p49 p78		1	
PUSH	stack pointer	3	3	p0156 p49 p78		1	
PUSHF(D/Q)		4	4	p06 p1 p49 p78		1	
PUSHA(D)		19	19			6	not 64 bit
POP	r	1	1	p23	3	0.5	
POP	stack pointer	3	3				
POP	m	3	3	p23 p49 p78		1	
POPF(D/Q)		9	9			21	
POPA(D)		18	18			8	not 64 bit
LAHF		1	1	p06	4	3	
SAHF		1	1	p06	1	1	
SALC		3	3		1	1	not 64 bit
LEA	r16,m	2	2		2	1	
LEA	r32/64,m	1	1	p0156	1	0.5	without index
LEA	r32/64,m	1	1	p15	1	0.5	with index
LEA	r32/64,m	1	1	p1		1	rip relative address
BSWAP	r32	1	1	p15	1	0.5	
BSWAP	r64	2	2	p0156	2	1	
MOVBE	r16,m16	3	3			0.5	
MOVBE	r32,m32	2	2			0.5	
MOVBE	r64,m64	3	3			0.5	
MOVBE	m16,r16	3	3			0.5	
MOVBE	m32,r32	3	3			0.5	
MOVBE	m64,r64	4	4			0.5	
PREFETCHNTA/0/1/2	m	1	1	p23		0.5	
PREFETCHW	m	1	1	p23		1	
LFENCE		2		none counted		5	
MFENCE		3	3			36	
SFENCE		2	2			6	
Arithmetic instructions							
ADD SUB	r,r/i	1	1	p0156	1	0.25	
ADD SUB	r,m	1	2	p0156 p23		0.5	
ADD SUB	m,r/i	2	4	p0156 p23 p49 p78	7	1	
ADC SBB	r,r/i	1	1	p06	1	1	
ADC SBB	r,m	1	2	p06 p23		1	
ADC SBB	m,r/i	4	4	p0156 p23 p49 p78	7	1	
ADCX ADOX	r,r	1	1	p06	1	1	
CMP	r,r/i	1	1	p0156	1	0.25	
CMP	m,r/i	1	2	p0156 p23	1	0.5	

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INC DEC NEG NOT	r	1	1	p0156	1	0.25	
INC DEC NOT	m	3	4		7	1	
NEG	m	2	4		7	1	
AAA AAS		2	2	p1 p0156	4		not 64 bit
DAA DAS		3	4		4		not 64 bit
AAD		3	3	p1 2p0156	6		not 64 bit
AAM		6	6		16		not 64 bit
MUL IMUL	r8	1	1	p1	3	1	
MUL IMUL	r16	4	4	p1 p0156	4	1	
MUL IMUL	r32	3	3	p1 2p0156	4	1	
MUL IMUL	r64	2	2	p1 p5	3	1	
IMUL	r,r	1	1	p1	3	1	
IMUL	r16,r16,i	2	2	p1 p0156	4	1	
IMUL	r32,r32,i	1	1	p1	3	1	
IMUL	r64,r64,i	1	1	p1	3	1	
MULX	r32,r32,r32	3	3	p1 2p056	4	1	
MULX	r64,r64,r64	2	2	p1 p5	4	1	
DIV IDIV	r8	4	4	p0156	19	6	
DIV IDIV	r16	4	4	p0156	17	6	
DIV IDIV	r32	4	4	p0156	12	6	
DIV IDIV	r64	4	4	p0156	15	10	
CBW		1	1	p0156	1	0.5	
CWDE		1	1	p0156	1	1	
CDQE		1	1	p0156	1	0.5	
CWD		2	2	p0156	1	1	
CDQ		1	1	p06	1	1	
CQO		1	1	p06	1	0.5	
POPCNT	r,r	1	1		3	1	
CRC32	r,r	1	1	p1	3	1	
Logic instructions							
AND OR XOR	r,r/i	1	1	p0156	1	0.25	
AND OR XOR	r,m	1	2	p0156 p23		0.5	
AND OR XOR	m,r/i	2	4	p0156 p23 p49 p78	7	1	
TEST	r,r/i	1	1	p0156	1	0.25	
TEST	m,r/i	1	2	p0156 p23	1	0.5	
SHR SHL SAR	r,i	1	1	p06	1	0.5	
SHR SHL SAR	m,i	4	4	p06 p23 p49 p78		2	
SHR SHL SAR	r,cl	2	2	2p06	1	1	
SHR SHL SAR	m,cl	5	6			4	
ROR ROL	r,1	2	2	2p06	1	1	short form
ROR ROL	r,i	1	1	p06	1	0.5	
ROR ROL	r,cl	2	2	2p06	1	1	
RCR RCL	r,1	3	3	2p06 p0156	2	2	
RCR	r,i	2	7		6	6	
RCL	r,i	2	7		7	7	
RCR	r,cl	2	7		6	6	
RCL	r,cl	2	7		7	7	
SHRD SHLD	r,r,i	1	1	p1	3	1	
SHLD	r,r,cl	3	3		3	1	
SHRD	r,r,cl	3	3		4	1	
SHLX SHRX SARX	r,r,r	1	1	p06	1	0.5	
RORX	r,r,i	1	1	p06	1	0.5	

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BT	r,r/i	1	1	p06	1	0.5	
BT	m,r	10	10			5	
BT	m,i	2	2	p06 p23		0.5	
BTR BTS BTC	r,r/i	1	1	p06	1	0.5	
BTR BTS BTC	m,r	10	11			5	
BTR BTS BTC	m,i	3	4			2	
BSF BSR	r,r	1	1	p1	3	1	
BSF BSR	r,m	1	2	p1 p23		1	
SETcc	r	1	1	p06	1	0.5	
SETcc	m	2	4			1	
CLC		1	0	none		0.25	
STC		1	1	p0156		0.25	
CMC		1	1	p0156	1		
CLD STD		3	3	p15 p6		4	
LZCNT	r,r	1	1	p1	3	1	
TZCNT	r,r	1	1	p1	3	1	
ANDN	r,r,r	1	1	p15	1	0.5	
BLSI BLSMSK	r,r	1	1	p15	1	0.5	
BLSR							
BEXTR	r,r,r	2	2	2p0156	2	0.5	
BEXTR	r,m,r	3	3	2p0156 p23		1	
BZHI	r,r,r	1	1	p15	1	0.5	
PDEP	r,r,r	1	1	p1	3	1	
PEXT	r,r,r	1	1	p1	3	1	
Control transfer instructions							
JMP	short/near	1	1	p6		1-2	
JMP	r	1	1	p6		2	
JMP	m	2	3			2	
Conditional jump	short/near	1	1	p06		1-2	predicted taken
Conditional jump	short/near	1	1	p06		0.5	predicted not taken
Fused arithmetic and branch		1	1	p06		1-2	predicted taken
Fused arithmetic and branch		1	1	p06		0.5	predicted not taken
J(E/R)CXZ	short	2	2	p0156 p6		0.5-2	
LOOP	short	7	7			4	
LOOP(N)E	short	4-12	4-12			6	
CALL	near	2	3			2	
CALL	r	2	3			3	
CALL	m	3	4			4	
RET		1	2			2	
RET	i		4			2	
BOUND	r,m	15	15			8	not 64 bit
INTO		5	5			6	not 64 bit
String instructions							
LODSB/W		3	3	2p0156 p23		1	
LODSD/Q		2	2	p0156 p23		1	
REP LODS		5n+12				~2n	
STOS		3	3	p0156 p49 p78		1	
REP STOS						20/64B	worst case

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REP STOS						1/64B	best case
MOVS		5	5			4	
REP MOVS						10/64B	worst case
REP MOVS						2/64B	best case
SCAS		3	3			1	
REP SCAS		~6n				~2n	
CMPS		5	5			4	
REP CMPS		~6n				~2n	
Synchronization instructions							
XADD	m,r	4	5			7	
LOCK XADD	m,r	9	9			21	
LOCK ADD	m,r	8	8			21	
CMPXCHG	m,r	5	6			7	
LOCK CMPXCHG	m,r	10	10			22	
CMPXCHG8B	m,r	16	16			11	
LOCK CMPXCHG8B	m,r	19	19			23	
CMPXCHG16B	m,r	23	23			16	
LOCK CMPXCHG16B	m,r	24	24			32	
Other							
NOP		1	0	none		0.2	
Long NOP		1	0	none		0.2	
PAUSE		6	6	p6		138	
ENTER	a,0	12	12			8	
ENTER	a,b	~13+8b			~100+2b		
LEAVE		3	3			4	
XGETBV		15	15			8	
CPUID		27-70				100-250	
RDTSC		21	21			27	
RDTSCP		22	22			36	
RDPMSR		20	20			9	
RDRAND	r	34	34			~2000	
RDSEED	r	35	35			~2000	

Floating point x87 instructions

Instruction	Operands	μops fused domain	μops unfused domain	μops each port	Latency	Reciprocal throughput	Comments
Move instructions							
FLD	r	1	1	p05	1	0.5	
FLD	m32/64	1	1	p23	4	0.5	
FLD	m80	4	4		4	2	
FBLD	m80	47	47		43	25	
FST(P)	r	1	1	p05	1	0.5	
FST(P)	m32/m64	1	2	p49 p78	3	1	
FSTP	m80	9	9		3		
FBSTP	m80	226	226		264	254	
FXCH	r	2	0	none	0	0.5	
FILD	m	1	2	p5 p23	6	1	
FIST(P)	m	3	4		5	1	
FISTTP	m	3	5			2	

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FLDZ		1	1	p5		1
FLD1		2	3			2
FLDPI FLDL2E etc.		2	3			2
FCMOVcc	r	4	4		3	2
FNSTSW	AX	2	2		4	1
FNSTSW	m16	2	3			1
FLDCW	m16	3	3			3
FNSTCW	m16	2	3			1
FINCSTP FDECSTP		1	1	p05	0	0.5
FFREE(P)	r	1	1	p05		0.5
FNSAVE	m	117	117		164	164
FRSTOR	m	80	80		181	181
Arithmetic instructions						
FADD(P)						
FSUB(R)(P)	r	1	1	p5	3	1
FADD(P)						
FSUB(R)(P)	m	2	3	p5 p23		1
FMUL(P)	r	1	1	p0	4	1
FMUL(P)	m	2	3	p0 p23		1
FDIV(R)(P)	r	1	1	p0	14-16	4-5
FDIV(R)(P)	m	1	3			4-5
FABS		1	1	p0	1	1
FCHS		1	1	p0	1	1
FCOM(P) FUCOM	r	1	1	p5	3	1
FCOM(P) FUCOM	m	1	2	p5 p23		1
FCOMPP FUCOMPP		2	2	p0 p5		1
FCOMI(P)						
FUCOMI(P)	r	3	3	p0	4	1
FIADD FISUB(R)	m	3	3	2p5 p23		2
FIMUL	m	3	4	p0 p5 p23		1
FIDIV(R)	m	2	3	p0 p5 p23		5
FICOM(P)	m	2	3	2p5 p23		2
FTST		1	1	p5	3	1
FXAM		2	2	2p5	6	2
FPREM		37	37		27-30	26
FPREM1		37	37		29-60	26
FRNDINT		17	17		21	11
Math						
FSCALE		29	29		11	11
FXTRACT		17	17		11	10
FSQRT		1	1	p0	14-21	4-7
FSIN			60-100		60-120	
FCOS			60-100		60-130	
FSINCOS			60-120		60-140	
F2XM1		16-90			60-80	
FYL2X			60-100		50-105	
FYL2XP1		56			60-70	
FPTAN		110	110		60-140	
FPATAN			30-100		100-160	
Other						

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FNOP	1	1	p05	0.5
WAIT	2	2	p05	1
FNCLEX	6	6		28
FNINIT	15	15		73

Integer vector instructions

Instruction	Operands	μops fused domain	μops unfused domain	μops each port	Latency	Reciprocal through put	Comments
Move instructions							
MOVD	r,v	1	1	p0	2	1	
MOVD	m,v	1	2	p49 p78	2	0.5	
MOVD	v,r	1	1	p5	2	1	
MOVD	v,m	1	1	p23	2	0.5	
MOVQ	r,v	1	1	p0	2	1	
MOVQ	v,r	1	1	p5	2	1	
MOVQ	mm,mm	1		p05	1	0.5	
MOVQ	x,x	1		p0156	1	0.33	
MOVQ	v,m	1	1	p23	3	0.5	
MOVQ	m,v	1	2	p49 p78	2	0.5	
MOVDQA/U	x,x	1	1		0	0.2	may eliminate
MOVDQA/U	x, m128	1	1	p23	2	0.5	
MOVDQA/U	m128, x	1	2	p49 p78	2	0.5	
VMOVDQA/U	y,y	1	1		0	0.2	may eliminate
VMOVDQA32/64	z,z	1	1		0	0.2	may eliminate
VMOVDQA32/64	xy{k},xy	1	1	p015	1	0.33	
VMOVDQA32/64	z{k},z	1	1	p05	1	0.5	
VMOVDQA/U/32/64	y,m	1	1	p23	4	0.5	
VMOVDQA/U/32/64	z,m	1	1	p23	4	0.5	
VMOVDQA/U32	v{k},m	1	2	p05 p23	5	1	
VMOVDQA/U32	v{k}{z},m	1	2	p05 p23	5	0.5	
VMOVDQA/U/32/64	m,y	1	2	p49 p78	3	0.5	
VMOVDQA/U/32/64	m,z	1	2	p49 p78	3	1	
VMOVDQA/U32	m{k},v	1	2		16	1	
LDDQU	x,m	1	1	p23	3	0.5	
MOVDQ2Q	mm, x	2	2	2p015	2	1	
MOVQ2DQ	x,mm	2	2	p0 p015	2	1	
MOVNTQ	m64,mm	1	2	p49 p78	~450	4	
MOVNTDQ	m128,x	1	2	p49 p78	~450	4	
VMOVNTDQ	m256,y	1	2	p49 p78	~450	4	
MOVNTDQA	v,m	2	2	p23 p015	5	0.5	
PACKSSWB/DW							
PACKUSWB	mm,mm	3	3	p5	4	2	
PACKSSWB/DW							
PACKUSWB	x,x / v,v,v	1	1	p5	1	1	
PACKUSDW	x,x / v,v,v	1	1	p5	1	1	
PUNPCKH/LBW	mm,mm	1	1	p5	1	1	
PUNPCKH/L...	v,v / v,v,v	1	1	p15	1	0.5	
PMOVSX/ZX BW BD							
BQ DW DQ	x,x	1	1	p15	1	0.5	

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VPMOVXS/ZX BW BD BQ DW DQ	y,x	1	1	p5	3	1	
VPMOVXS/ZX BW BD BQ DW DQ	z,xy	1	1	p5	3	1	
VPMOVWB/DB/QB/ DW/QW/QD WPMOVSWB/DB/QB/ DW/QW/QD WPMOVUSWB/DB/QB/ /DW/QW/QD	v,v	2	2	2p5	4	2	
PSHUFB	mm,mm	3	3	p0 p5	3	1	
PSHUFB	x,x / y,y,y	1	1	p15	1	0.5	
VPSHUFB	z,z,z	1	1	p5	1	1	
PSHUFW	mm,mm,i	1	1	p5	1	1	
PSHUFL/HW	x,x,i / y,y,i	1	1	p15	1	0.5	
PSHUFL/HW	z,z,i	1	1	p5	1	1	
PSHUFD	x,x,i / y,y,i	1	1	p15	1	0.5	
PSHUFD	z,z,i	1	1	p5	1	1	
VSHUFI32X4	v,v,v,i	1	1	p5	3	1	
VSHUFI64X2	v,v,v,i	1	1	p5	3	1	
VPSHUFBITQMB	k,v,v	2	2	p0 p1	8	1	avx512bitalg
PALIGNR	v,v,i / v,v,v,i	1	1	p5	1	1	
VALIGND/Q	v,v,v,i	1	1	p5	3	1	
PBLENDVB	x,x,xmm0	1	1	p015	1	0.5	
VPBLENDVB	v,v,v,v	2	2	2p015	2	1	
PBLENDW	x,x,i / v,v,v,i	1	1	p15	1	0.5	
VPBLEND	v,v,v,i	1	1	p015	1	0.33	
VPBLENDMB/W	v{k},v,v	1	1	p05	3	0.5	
VPBLENDMD/Q	v{k},v,v	1	1	p05	1	0.5	
VPERMW	v,v,v	2	2	p0 p5	3	1	
VPERMD	v,v,v	1	1	p5	3	1	
VPERMQ	v,v,i	1	1	p5	3	1	
VPERMQ	v,v,v	1	1	p5	3	1	
VPERM2I128	y,y,y,i	1	1	p5	3	1	
VPERMI2W	v,v,v	3	3	p01 2p5	7	2	
VPERMI2D	v,v,v	1	1	p5	3	1	
VPERMI2Q	v,v,v	1	1	p5	3	1	
VPERMT2W	v,v,v	3	3	p01 2p5	6	2	
VPERMT2D	v,v,v	1	1	p5	3	1	
VPERMT2Q	v,v,v	1	1	p5	3	1	
MASKMOVQ	mm,mm	4	4	p0 p4 2p23	~400	4	
MASKMOVDQU	x,x	10	10	4p04 2p56 4p23	~450	9	
VPMASKMOVD/Q	v,v,m	2	2	p23 p015	4	0.5	
VPMASKMOVD/Q	m,v,v	3	3	p0 p4 p23	4	1	
PMOVMSKB	r,v	1	1	p0	2	1	
PEXTRB/W/D/Q	r,x,i	2	2	p0 p15	3	1	
PEXTRB/W/D/Q	m,x,i	2	3	p15 p49 p78	3	0.5	
VEXTRACTI128	x,y,i	1	1	p5	3	1	
VEXTRACTI128	m,y,i	2	2	p49 p78	4	0.5	
PINSRB/W/D/Q	x,r32,i	2	2	2p15	3	1	
PINSRB/W/D/Q	x,m,i	2	2	p23 p15	3	1	
VINSERTI128	y,y,x,i	1	1	p5	3	1	
VINSERTI128	y,y,m,i	2	2	p015 p23	4-5	0.5	
VINSERTI32x4	z,z,x,i	1	1	p5	3	1	
VINSERTI32x4	z,z,m,i	1	2	p05 p23	4-5	0.5	

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VINSERTI64x4	z,z,y,i	1	1	p5	3	1	
VINSERTI64x4	z,z,m,i	1	2	p05 p23	4-5	0.5	
VPBROADCAST B/W/D/Q	x,x	1	1	p5	1	1	
VPBROADCAST B/W	v,m	2	2	p23 p5		1	
VPBROADCAST D/Q	v,m	1	1	p23		0.5	
VPBROADCAST B/W/D/Q	v,x	1	1	p5	3	1	
VBROADCASTI128	y,m128	1	1	p23	4	0.5	
VBROADCASTI32X2	z,x,i	1	1	p5	3	1	
VBROADCASTI32X4/ i64x2	z,m128,i	2	2	p05 p23	4	0.5	
VBROADCASTI32X8	z,y,i	1	1	p5	3	1	
VBROADCASTI32X8/ i64x4	z,m256,i	1	1	p23	4	0.5	
VPCOMPRESSB/W	v{k},v	2	2	p5	3		AVX512VBMI2
VPCOMPRESSD/Q	v{k},v	2	2	p5	3		
VPEXPANDB/W	v{k},v	2	2	p5	5		AVX512VBMI2
VPEXPANDD/Q	v{k},v	2	2	p5	3		
VPGATHERDD	x,[r+s*x],x	4	4	p015 p23		3	
VPGATHERDD	y,[r+s*y],y	4	4	p015 p23		5	
VPGATHERDQ	x,[r+s*x],x	4	5	p015 p23		1	
VPGATHERDQ	y,[r+s*x],y	4	4	p015 p23		3	
VPGATHERQD	x,[r+s*x],x	4	5	p015 p23		1	
VPGATHERQD	x,[r+s*y],x	4	4	p015 p23		3	
VPGATHERQQ	x,[r+s*x],x	4	5	p015 p23		1	
VPGATHERQQ	y,[r+s*y],y	4	4	p015 p23		3	
VPGATHERDD	x{k},[r+s*x]	4	4	p015 p23		3	
VPGATHERDD	y{k},[r+s*y]	4	4	p015 p23		5	
VPGATHERDD	z{k},[r+s*z]	4	4	p05 p23		9	
VPGATHERDQ	x{k},[r+s*x]	4	5	p015 p23		2	
VPGATHERDQ	y{k},[r+s*x]	4	4	p015 p23		3	
VPGATHERDQ	z{k},[r+s*y]	4	4	p05 p23		5	
VPGATHERQD	x{k},[r+s*x]	4	5	p015 p23		2	
VPGATHERQD	x{k},[r+s*y]	4	4	p015 p23		3	
VPGATHERQD	y{k},[r+s*z]	4	4	p05 p23		5	
VPGATHERQQ	x{k},[r+s*x]	4	5	p015 p23		2	
VPGATHERQQ	y{k},[r+s*y]	4	4	p015 p23		3	
VPGATHERQQ	z{k},[r+s*z]	4	4	p05 p23		5	
VPSCATTERDD	[r+s*x]{k},x	12	12			6	
VPSCATTERDD	[r+s*y]{k},y	20	20			8	
VPSCATTERDD	[r+s*z]{k},z	36	36			12	
VPSCATTERDQ	[r+s*x]{k},x	8	8			5	
VPSCATTERDQ	[r+s*x]{k},y	12	12			6	
VPSCATTERDQ	[r+s*y]{k},z	20	20			8	
VPSCATTERQD	[r+s*x]{k},x	8	8			5	
VPSCATTERQD	[r+s*y]{k},x	12	12			6	
VPSCATTERQD	[r+s*z]{k},y	20	20			8	
VPSCATTERQQ	[r+s*x]{k},x	8	8			5	
VPSCATTERQQ	[r+s*y]{k},y	12	12			6	
VPSCATTERQQ	[r+s*z]{k},z	20	20			8	

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Arithmetic instructions							
PADD/SUB(S,US) B/W/D/Q	mm,mm	1	1	p05	1	0.5	
PADD/SUB(S,US) B/W/D/Q	x,x / xy,xy,xy	1	1	p015	1	0.33	
PADD/SUB(S,US) B/W/D/Q	z,z,z	1	1	p05	1	0.5	
PADD/SUB(S,US) B/W/D/Q	v,m / v,v,m	1	2	p015 p23		0.5	
PHADD(S)W/D							
PHSUB(S)W/D	v,v / v,v,v	3	3	p015	2	1	
PCMPEQB/W/D							
PCMPGTB/W/D	mm,mm	1	1	p0	1	1	
PCMPEQB/W/D							
PCMPGTB/W/D	x,x / y,y,y	1	1	p01	1	0.5	
PCMPEQQ	v,v / v,v,v	1	1	p01	1	0.5	
PCMPGTQ	v,v / v,v,v	1	1	p5	3	1	
VPCMP.B/W/D/Q	k{k},v,v	1	1	p5	3	1	
PMULL/HW							
PMULHUW	mm,mm	1	1	p0	5	1	
PMULL/HW							
PMULHUW	x,x / y,y,y	1	1	p01	5	0.5	
VPMULL/HW							
VPMULHUW	z,z,z	1	1	p0	5	1	
VPMULLQ	xy,xy,xy	3	3	p01	15	1.5	
VPMULLQ	v,v,v	3	3	p0	15	3	
PMULHRSW	mm,mm	1	1	p0	5	1	
PMULHRSW	x,x / y,y,y	1	1	p01	5	0.5	
PMULHRSW	z,z,z	1	1	p0	5	1	
PMULLD	x,x / y,y,y	2	2	2p01	10	1	
PMULLD	z,z,z	2	2	2p0	10	2	
PMULDQ	x,x / y,y,y	1	1	p01	5	0.5	
PMULDQ	z,z,z	1	1	p0	5	1	
PMULUDQ	mm,mm	1	1	p0	5	1	
PMULUDQ	x,x / y,y,y	1	1	p01	5	0.5	
PMULUDQ	z,z,z	1	1	p0	5	1	
PMADDWD	mm,mm	1	1	p0	5	1	
PMADDWD	x,x / y,y,y	1	1	p01	5	0.5	
PMADDWD	z,z,z	1	1	p0	5	1	
PMADDUBSW	mm,mm	1	1	p0	5	1	
PMADDUBSW	x,x / y,y,y	1	1	p01	5	0.5	
PMADDUBSW	z,z,z	1	1	p0	5	1	
VPDPBUSD	v,v,v	1	1	p01	5	0.5-1	avx512vnni
VPDPBUSDS	v,v,v	1	1	p01	5	0.5-1	avx512vnni
VPDPWSSD	v,v,v	1	1	p01	5	0.5-1	avx512vnni
VPDPWSSDS	v,v,v	1	1	p01	5	0.5-1	avx512vnni
VPMADD52HUQ	v,v,v	1	1	p01	4	0.5-1	avx512ifma
VPMADD52LUQ	v,v,v	1	1	p01	4	0.5-1	avx512ifma
PAVGB/W	mm,mm	1	1	p0	1	1	
PAVGB/W	x,x / y,y,y	1	1	p01	1	0.5	
PAVGB/W	z,z,z	1	1	p0	1	1	
PMIN/PMAX							
SB/SW/SD							
UB/UW/UD	mm,mm	1	1	p0	1	1	

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PMIN/PMAX SB/SW/SD UB/UW/UD	x,x / y,y,y	1	1	p01	1	0.5	avx512bitalg
PMIN/PMAX SB/SW/SD UB/UW/UD	z,z,z	1	1	p0	1	1	
PHMINPOSUW	x,x	1	1	p0	4	1	
PABSB/W/D	mm,mm	1	1	p0	1	1	
PABSB/W/D	x,x / y,y	1	1	p01	1	0.5	
PABSB/W/D	z,z,z	1	1	p0	1	1	
PSIGNB/W/D	mm,mm	1	1	p0	1	1	
PSIGNB/W/D	x,x / y,y,y	1	1	p01	1	0.5	
PSADBW	v,v / v,v,v	1	1	p5	3	1	
MPSADBW	x,x,i / v,v,v,i	2	2	p1 p5	5	1	
VPOPCNTB/W/D/Q	v,v	1	1	p5	3	1	
Logic instructions							
PAND PANDN POR PXOR	mm,mm	1	1	p05	1	0.5	
PAND PANDN POR PXOR	x,x / y,y,y	1	1	p015	1	0.33	
VPAND/ANDN/OR/ XOR D/Q	z,z,z	1	1	p05	1	0.5	
PAND PANDN POR PXOR	v,m / v,v,m	1	2	p015 p23		0.5	
PTEST	x,x	2	2	p0 p5	3	1	
VPTEST	y,y	2	2	p0 p5	5	1	
VPTESTMB/W/D/Q VPTESTNMB/W/D/Q	k,z,z	1	1	p5	3	1	
PSLLW/D/Q PSRLW/D/Q	mm,mm	1	1	p0	1	1	
PSLLW/D/Q PSRLW/D/Q	x,x / v,v,x	2	2	p015	1	1	
PSLLW/D/Q PSRLW/D/Q	mm,i	1	1	p0	1	1	
PSLLW/D/Q PSRLW/D/Q	x,i / y,y,i	1	1	p01	1	0.5	
VPSLLW/D/Q VPSRLW/D/Q	z,z,i	1	1	p0	1	1	
VPSLLW/D/Q VPSRLW/D/Q	v,v,x	2	2	p015	2	1	
VPSLLVD/Q VPSRAVD	xy,xy,xy	1	1	p01	1	0.5	
VPSLLVW/D/Q VPSRAVW/D/Q	z,z,z	1	1	p0	1	1	
PSLLDQ PSRLDQ	x,i / y,y,i	1	1	p15	1	0.5	
PSLLDQ PSRLDQ	z,z,i	1	1	p5	1	1	

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VPSHLDW/D/Q	v,v,v,i	1	1	p01	1	0.5-1	avx512vbmi2
VPSHRDW/D/Q	v,v,v,i	1	1	p01	1	0.5-1	avx512vbmi2
VPSHLDVW/D/Q	v,v,v						not measured
VPSHRDVW/D/Q	v,v,v						not measured
VPMULTISHIFTQB	v,v,v	1	1	p5	3	1	avx512vbmi
VPROL/RD/Q	xy,xy,i	1	1	p01	1	0.5	
VPROL/RD/Q	z,z,i	1	1	p0	1	1	
VPROL/RVD/Q	xy,xy,xy	1	1	p01	1	0.5	
VPROL/RVD/Q	z,z,z	1	1	p0	1	1	
VPLZCNTD/Q	xy,xy	1	1	p01	4	0.5	
VPLZCNTD/Q	z,z	1	1	p0	4	1	
VPTERNLOGD/Q	v,v,v	1	1	p05	1	0.5	
String instructions							
PCMPESTRI	x,x,i	8	8		11	4	
PCMPESTRI	x,m128,i	8	8			4	
PCMPESTRM	x,x,i	9	9		11	5	
PCMPESTRM	x,m128,i	9	9			5	
PCMPISTRI	x,x,i	3	3	3p0	10	3	
PCMPISTRI	x,m128,i	4	4			3	
PCMPISTRM	x,x,i	3	3	3p0	9	3	
PCMPISTRM	x,m128,i	4	4			3	
Encryption instructions							
PCLMULQDQ	x,x,i	1	1	p5	6	1	pclmulqdq
VPCLMULQDQ	y,y,y,i	3	3	p01 2p5	8	2	vpcmulqdq
AESDEC, AESDECLAST, AESENC, AESENCLAST	x,x	1	1	p01	3	0.5	aes
AESIMC	x,x	2	2	2p01	6	1	aes
AESKEYGENASSIST	x,x,i	14	14		12	12	aes
VAESDEC, VAESDECLAST, VAEENC, VAESENCLAST	y,y,y	1	1	p01	3	0.5	vaes
SHA1RND54	x,x,i	1	1	p5	6	3	sha
SHA1NEXTE	x,x	3	3	p0 p1 p5	2	1	sha
SHA1MSG1	x,x	2	2	p01 p5	2	1	sha
SHA1MSG2	x,x	6	6	4p01 2p5	6	4	sha
SHA256RND52	x,x,xmm0	1	1	p5	6	3	sha
SHA256MSG1	x,x	6	6	4p01 2p5	5	4	sha
SHA256MSG2	x,x	12	12	8p01 4p5	11	5	sha
Other							
EMMS		10	10	p05 p6		7	
VG2P8AFFINEINVBQ	v,v,v,i	1	1	p01	3	0.5-1	avx512gfni
VG2P8AFFINEVBQ	v,v,v,i	1	1	p01	3	0.5-1	avx512gfni
VG2P8MULB	v,v,v	1	1	p01	3	0.5-1	avx512gfni
VPCONFLICTD	x,x	15	15	p015	12	6	avx512cd
VPCONFLICTD	y,y	22	22	p015	17	9	avx512cd
VPCONFLICTD	z,z	37	37	p05	26	19	avx512cd
VPCONFLICTQ	x,x	3	3	p015	4	2	avx512cd

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VPCONFLICTQ	y,y	15	15	p015	13	6	avx512cd
VPCONFLICTQ	z,z	22	22	p05	17	11	avx512cd
VP2INTERSECTD/Q	k+1,v,v			not tested			AVX512VP2IN TERSECT

Floating point vector instructions

Instruction	Operands	μops fused domain	μops unfused domain	μops each port	Latency	Recipro- cal through put	Comments
Move instructions							
MOVAPS/D	v,v	1	1		0	0.2	may eliminate
VMOVAPS/D	xy{k},xy	1	1	p015	1	0.33	
VMOVAPS/D	z{k},z	1	1	p05	1	0.5	
VMOVAPS/D							
VMOVUPS/D	v,m	1	1	p23	3-4	0.5	
VMOVAPS/D							
VMOVUPS/D	xy{k},m	1	2	p23	4-5	0.5	
VMOVAPS/D							
VMOVUPS/D	z{k},m	1	2	p23	4-5	1	
VMOVAPS/D							
VMOVUPS/D	z{k}{z},m	1	2	p23	4-5	0.5	
MOVAPS/D							
MOVUPS/D	m,xy	1	2	p49 p78	3	0.5	
VMOVAPS/D							
VMOVUPS/D	m,z	1	2	p49 p78	3	1	
VMOVAPS/D							
VMOVUPS/D	m{k},z	1	2	p49 p78	16	1	
MOVSS/D	x,x	1	1	p015	1	0.33	
MOVSS/D	x,m	1	1	p23	4	0.5	
MOVSS/D	m,x	1	2	p49 p78	3	0.5	
MOVHPS/D	x,m64	1	2	p23 p5	4	1	
MOVHPS/D	m64,x	1	2	p49 p78	3	0.5	
MOVLPS/D	x,m64	1	2	p23 p15	4	0.5	
MOVLPS/D	m64,x	1	2	p49 p78	3	0.5	
MOVHLPs	x,x	1	1	p5	1	1	
MOVLHPS	x,x	1	1	p5	1	1	
MOVMSKPS/D	r32,x	1	1	p0	2	1	
VMOVMSKPS/D	r32,y	1	1	p0	4	1	
MOVNTPS/D	m128,x	1	2	p40 p78	~450	4	
VMOVNTPS/D	m256,y	1	2	p40 p78	~450	4	
SHUFPS/D	x,x,i / v,v,v,i	1	1	p15	1	0.5	
VSHUFF32X4	v,v,v,i	1	1	p5	3	1	
VSHUFF64X2	v,v,v,i	1	1	p5	3	1	
VPERMILPS/PD	v,v,i	1	1	p5	1	1	
VPERMILPS/PD	v,v,v	1	1	p5	1	1	
VPERM2F128	y,y,y,i	1	1	p5	3	1	
VPERMPS/D	v,v,v	1	1	p5	3	1	
VPERMPD	y,y,i	1	1	p5	3	1	
VPERMI2PS/PD	v,v,v	1	1	p5	3	1	
VPERMT2PS/PD	v,v,v	1	1	p5	3	1	
BLENDPS/PD	x,x,i / y,y,y,i	1	1	p015	1	0.33	
BLENDDPS/PD	x,x,xmm0	1	1	p015	1	0.5	

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VBLENDVPS/PD	v,v,v,v	2	2	2p015	2	1
VBLENDMPS/PD	v{k},v,v	1	1	p05	1	0.5
MOVDDUP	v,v	1	1	p5	1	1
MOVDDUP	v,m	1	1	p23	3	0.5
VBROADCASTSS/D	v,m	1	1	p23	3-4	0.5
VBROADCASTSS/D	x,x	1	1	p5	1	1
VBROADCASTSS/D	yz,x	1	1	p5	3	1
VBROADCASTF128	y,m128	1	1	p23	5	0.5
MOVSH/LDUP	xy,xy	1	1	p15	1	0.5
MOVSH/LDUP	z,z	1	1	p5	1	1
UNPCKH/LPS/D	x,x / v,v,v	1	1	p5	1	1
EXTRACTPS	r32,x,i	2	2	p0 p5	3	1
EXTRACTPS	m32,x,i	2	3	p5 p49 p78	5	1
VEEXTRACTF128	x,y,i	1	1	p5	3	1
VEEXTRACTF128	m128,y,i	2	2	p49 p78	5	0.5
VEEXTRACTF32X4						
VEEXTRACTF64X2	x,z,i	1	1	p5	3	1
VEEXTRACTF32X4						
VEEXTRACTF64X2	m128,z,i	2	2	p49 p78	4	1
INSERTPS	x,x,i	1	1	p5	1	1
INSERTPS	x,m32,i	2	2	p23 p5	4	1
VINSERTF128	y,y,x,i	1	1	p5	3	1
VINSERTF128	y,y,m128,i	2	2	p015 p23	5	0.5
VEEXTRACTF32X4						
VEEXTRACTF64X2	z,z,x,i	1	1	p5	3	1
VEEXTRACTF32X4						
VEEXTRACTF64X2	z,z,m128,i	2	2	p05 p23	4	1
VMASKMOVPS/D	v,v,m	2	2	p015 p23	3	0.5
VMASKMOVPS/D	m,xy,xy	3	3		16	1
VCOMPRESSPS/PD	v{k},v	2	2	p5	3	
VEXPANDPS/PD	v{k},v	2	2	p5	3	
VGATHERDPS	x,[r+s*x],x	4	4	p015 p23		3
VGATHERDPS	y,[r+s*y],y	4	4	p015 p23		5
VGATHERDPD	x,[r+s*x],x	4	5	p015 p23		1
VGATHERDPD	y,[r+s*x],y	4	4	p015 p23		3
VGATHERQPS	x,[r+s*x],x	4	5	p015 p23		1
VGATHERQPS	x,[r+s*y],x	4	4	p015 p23		3
VGATHERQPD	x,[r+s*x],x	4	5	p015 p23		1
VGATHERQPD	y,[r+s*y],y	4	4	p015 p23		3
VGATHERDPS	x{k},[r+s*x]	4	4	p015 p23		3
VGATHERDPS	y{k},[r+s*y]	4	4	p015 p23		5
VGATHERDPS	z{k},[r+s*z]	4	4	p05 p23		9
VGATHERDPD	x{k},[r+s*x]	4	5	p015 p23		2
VGATHERDPD	y{k},[r+s*x]	4	4	p015 p23		3
VGATHERDPD	z{k},[r+s*y]	4	4	p05 p23		5
VGATHERQPS	x{k},[r+s*x]	4	5	p015 p23		2
VGATHERQPS	x{k},[r+s*y]	4	4	p015 p23		3
VGATHERQPS	y{k},[r+s*z]	4	4	p05 p23		5
VGATHERQPD	x{k},[r+s*x]	4	5	p015 p23		2
VGATHERQPD	y{k},[r+s*y]	4	4	p015 p23		3
VGATHERQPD	z{k},[r+s*z]	4	4	p05 p23		5
VSCATTERDPS	[r+s*x]{k},x	12	12			6
VSCATTERDPS	[r+s*y]{k},y	20	20			8
VSCATTERDPS	[r+s*z]{k},z	36	36			12

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VSCATTERDPD	[r+s*x]{k},x	8	8			5	
VSCATTERDPD	[r+s*x]{k},y	12	12			6	
VSCATTERDPD	[r+s*y]{k},z	20	20			8	
VSCATTERQPS	[r+s*x]{k},x	8	8			5	
VSCATTERQPS	[r+s*y]{k},y	12	12			6	
VSCATTERQPS	[r+s*z]{k},z	20	20			8	
VSCATTERQPD	[r+s*x]{k},x	8	8			5	
VSCATTERQPD	[r+s*y]{k},y	12	12			6	
VSCATTERQPD	[r+s*z]{k},z	20	20			8	
Conversion							
CVTPD2PS	x,x	2	2	p01 p5	5	1	
VCVTPD2PS	x,y / y,z	2	2	p01 p5	7	1	
CVTPS2PD	x,x	2	2	p01 p5	5	1	
VCVTPS2PD	y,x / z,y	2	2	p01 p5	7	1	
CVTSS2SD	x,x	2	2	p01 p5	5	2	
CVTSD2SS	x,x	2	2	p01 p5	5	1	
CVT(T) PS2DQ	x,x	1	1	p01	4	0.5	
VCVT(T) PS2DQ	y,y	1	1	p01	4	0.5	
VCVT(T) PS2DQ	z,z	1	1	p0	4	1	
CVTDQ2PS	x,x	1	1	p01	4	0.5	
VCVTDQ2PS	y,y	1	1	p01	4	0.5	
VCVTDQ2PS	z,z	1	1	p0	4	1	
VCVT(T)PS2QQ	v,v	2	2	p01 p5	5	1	
VCVTQQ2PS	v,v	2	2	p01 p5	5-7	1	
VCVTPS2QQ	v,v	2	2	p01 p5	5-7	1	
VCVT(T)PS2UDQ	v,v	1	1	p01	4	0.5-1	
VCVTUDQ2PS	v,v	1	1	p01	4	0.5-1	
VCVT(T)PS2UQQ	v,v	2	2	p01 p5	5-7	1	
VCVTUQQ2PS	v,v	2	2	p01 p5	5-7	1	
CVT(T)PS2PI	mm,x	2	2	p0 p15	6	1	
CVTPi2PS	x,mm	2	2	p0 p01	7	2	
CVT(T)SS2SI	r32,x	2	2	2p01	6	1	
CVT(T)SS2SI	r64,x	3	3	2p01 p5	7	1	
CVTSI2SS	x,r32	2	2	p01 p5	6	2	
CVTSI2SS	x,r64	3	3	p01 2p5	7	2	
CVT(T)SS2USI	r32,x	2	2	2p01	6	1	
CVT(T)SS2USI	r64,x	3	3	3p015	7	1	
CVTUSI2SS	x,r32	2	2	p01 p5	6	1	
CVTUSI2SS	x,r64	3	3	p01 2p5	7	2	
VCVTPS2PH	x,x,i	2	2	p01 p5	5	1	
VCVTPS2PH	x,y,i	2	2	p01 p5	7	1	
VCVTPS2PH	y,z,i	2	2	p0 p5	7	1	
VCVTPH2PS	x,x	2	2	p01 p5	5	1	
VCVTPH2PS	y,x	2	2	p01 p5	7	1	
VCVTPH2PS	z,y	2	2	p0 p5	7	1	
CVT(T)PD2DQ	x,x	2	2	p01 p5	5	1	
VCVT(T)PD2DQ	x,y	2	2	p01 p5	7	1	
VCVT(T)PD2DQ	y,z	2	2	p0 p5	7	1	
CVTDQ2PD	x,x	2	2	p01 p5	5	1	
VCVTDQ2PD	y,x	2	2	p01 p5	7	1	
VCVTDQ2PD	z,y	2	2	p0 p5	7	1	
VCVT(T)PD2QQ	v,v	1	1	p01	4	0.5-1	

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VCVTQQ2PD	v,v	1	1	p01	4	0.5-1
VCVT(T)PD2UDQ	x,x	2	2	p01 p5	5	1
VCVT(T)PD2UDQ	x,y	2	2	p01 p5	7	1
VCVT(T)PD2UDQ	y,z	2	2	p0 p5	7	1
VCVTUDQ2PD	x,x	2	2	p01 p5	5	1
VCVTUDQ2PD	y,x	2	2	p01 p5	7	1
VCVTUDQ2PD	z,y	2	2	p0 p5	7	1
VCVT(T)PD2UQQ	v,v	1	1	p01	4	0.5-1
VCVTUQQ2PD	v,v	1	1	p01	4	0.5-1
CVT(T)PD2PI	mm,x	2	2	p01 p5	5	1
CVTPI2PD	x,mm	2	2	p01 p5	5	1
CVT(T)SD2SI	r32/64,x	2	2	p0 p1	6	1
CVTSI2SD	x,r32/64	2	2	p01 p5	6	2
CVT(T)SD2USI	r32/64,x	2	2	p0 p1	6	1
CVTUSI2SD	x,r32/64	2	2	p01 p5	6	1
Arithmetic						
ADDSS/D PS/D						
SUBSS/D PS/D	x,x / y,y,y	1	1	p01	4	0.5
VADDPS/D						
VSUBPS/D	z,z,z	1	1	p0	4	1
ADDSS/D PS/D						
SUBSS/D PS/D	x,m / v,v,m	1	2	p01 p23		0.5-1
ADDSUBPS/D	x,x / v,v,v	1	1	p01	4	0.5-1
HADDPS/D						
HSUBPS/D	x,x / v,v,v	3	3	p01 2p5	6	2
MULSS/D PS/D	x,x / v,v,v	1	1	p01	4	0.5
VMULPS/D	z,z,z	1	1	p0	4	1
MULSS/D PS/D	x,m / v,v,m	1	2	p01 p23		0.5-1
DIVSS	x,x	1	1	p0	11	3
DIVPS	xy,xy	1	1	p0	11	3
VDIVPS	z,z,z	3	3	2p0 p5	17	10
DIVSD	x,x	1	1	p0	13-14	4
DIVPD	x,x	1	1	p0	13-14	4
VDIVPD	y,y,y	1	1	p0	13-14	8
VDIVPD	z,z,z	3	3	2p0 p5	22-23	16
RCPSS/PS	v,v	1	1	p0	4	1
VRCP14SS/PS	xy,xy	1	1	p0	4	1
VRCP14PS	z,z	3	3	2p0 p5	6	2
CMPccSS/D						
CMPccPS/D	x,x / v,v,v	1	1	p01	4	0.5
VCMPPS/PD	k,v,v,i	1	1	p5	3	1
(U)COMISS/D	x,x	1	1	p0	2	1
MAXSS/SD/PS/PD						
MINSS/SD/PS/PD	x,x / v,v,v	1	1	p01	4	0.5-1
VRANGESS/SD/PS/PD	v,v,v	1	1	p01	4	0.5-1
ROUNDSS/SD/PS/PD	v,v,i	2	2	2p01	8	1
VRNDSCALE						
SS/PS/SD/PD	v,v,v,i	2	2	2p01	8	1
VREDUCE						
SS/PS/SD/PD	v,v,v,i	1	1	p01	4	0.5-1
DPPS	x,x,i / y,y,y,i	4	4	6p0156	14	4
DPPD	x,x,i	3	3	2p01 p5	9	1
VDBPSADBW	v,v,v,i	1	1	p5	3	1

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VFIXUPIMMSS/SD	v,v,v,i	1	1	p01	4	2	
VFPCLASSSS/SD	k,v,i	1	1	p5	3	1	
VFMADD... (all FMA instruct.)	xy,xy,xy	1	1	p01	4	0.5	
VFMADD... (all FMA instruct.)	z,z,z	1	1	p0	4	1	
Math							
SQRTSS/PS	x,x	1	1	p0	12	3	
VSQRTPS	y,y	1	1	p0	12	6	
VSQRTPS	z,z	3	3	2p0 p5	20	12	
SQRTSD	x,x	1	1	p0	15-16	4-6	
SQRTPD	x,x	1	1	p0	15-16	4-6	
VSQRTPD	y,y	1	1	p0	15-16	9-12	
VSQRTPD	z,z	3	3	2p0 p5	27-28	18-24	
RSQRTSS/PS	x,x	1	1	p0	4	1	
VRSQRTPS	y,y	1	1	p0	2	1	
VRSQRT14SS/PS	x,x	1	1	p0	4	1	
VRSQRT14PS	y,y	1	1	p0	2	1	
VRSQRT14PS	z,z	3	3	2p0 p5	4	2	
VGETEXPSS/SD	v,v,v	1	1	p01	4	0.5-1	
VGETMANTPS/PD	v,v,v	1	1	p01	4	0.5-1	
VSCALEFSS/D/PS/D	v,v,v	1	1	p01	4	0.5-1	
Logic							
AND/ANDN/OR/ XORPS/PD	x,x / y,y,y	1	1	p015	1	0.33	
AND/ANDN/OR/ XORPS/PD	z,z,z	1	1	p05	1	0.5	
AND/ANDN/OR/ XORPS/PD	z,z,m	1	2	p015 p23		0.5	
VTESTPS/PD	v,v	1	1	p0		1	
Other							
VZEROUPPER		4	4	none		1	
VZEROALL		21	21	p0 p1 p5 p6		12	32 bit mode
VZEROALL		29	29	p0 p1 p5 p6		11	64 bit mode
LDMXCSR	m32	4	4	p0 p5 p6 p23	6	3	
STMXCSR	m32	3	4		5	1	
FXSAVE	m	82	96		48	48	32 bit mode
FXSAVE	m	90	112		49	49	64 bit mode
FXRSTOR	m	97	97		74	74	32 bit mode
FXRSTOR	m	113	113		74	74	64 bit mode
XSAVE	m	271	303		131	131	32 bit mode
XSAVE	m	336	400		137	137	64 bit mode
XRSTOR	m	184	184		80	80	32 bit mode
XRSTOR	m	184	184		80	80	64 bit mode
XSAVEC	m	191	198		84	84	64 bit mode
XSAVEOPT	m	206	214		92	92	

Mask register instructions

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Instruction	Operands	μops fused domain	μops unfused domain	μops each port	Latency	Recipro- cal through put	Comments
Move instructions							
KMOVB/W/D/Q	k,k	1	1	p0	1	1	
KMOVB/W/D/Q	k,m	3	3	p23 p5	4	1	
KMOVB/W/D/Q	m,k	1	2	p49 p78	4	0.5	
KMOVB/W/D/Q	k,r	1	1	p5	2	1	
KMOVB/W/D/Q	r,k	1	1	p0	2	1	
VPBROADCASTMB2Q	v,k	2	2	p0 p5	7	1	
VPBROADCASTMW2D	v,k	2	2	p0 p5	7	1	
KUNPCKBW	k,k,k	1	1	p5	4	1	
KUNPCKWD	k,k,k	1	1	p5	4	1	
KUNPCKDQ	k,k,k	1	1	p5	4	1	
VPMOVM2B/W	v,k	1	1	p05	3	0.5	
VPMOVM2D/Q	v,k	1	1	p05	2	0.5	
VPMOVB/W2M	k,v	1	1	p0	3	1	
VPMOVD/Q2M	k,v	1	1	p0	2	1	
Arithmetic instructions							
KADDB/W/D/Q	k,k,k	1	1	p5	4	1	
KSHIFTLB/W/D/Q	k,k,i	1	1	p5	4	1	
KSHIFTRB/W/D/Q	k,k,i	1	1	p5	4	1	
Logic instructions							
KANDB/W/D/Q	k,k,k	1	1	p0	1	1	
KANDNB/W/D/Q	k,k,k	1	1	p0	1	1	
KORB/W/D/Q	k,k,k	1	1	p0	1	1	
KXORB/W/D/Q	k,k,k	1	1	p0	1	1	
KXNORB/W/D/Q	k,k,k	1	1	p0	1	1	
KNOTB/W/D/Q	k,k	1	1	p0	1	1	
KTESTB/W/D/Q	k,k	1	1	p0	2	1	
KORTESTB/W/D/Q	k,k	1	1	p0	2	1	